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Roll No

BE-3001 (EE/EI/EX)-CBGS

B.E., III Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Mathematics - III

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Find the Fourier series for $f(x) = e^{-x}$ in $0 < x < 2\pi$.
b) Find the half range cosine series for the function $f(x) = x$ in $0 < x < \pi$.

2. a) Find the Fourier complex transform of

$$f(x) = \begin{cases} x^2, & |x| < a \\ 0, & |x| > a \end{cases}$$

- b) Find the Fourier cosine transform of $f(x) = e^{-3x} + e^{-4x}$.

3. a) Find $L\{1 + t^{3/2} - 3e^{-2t} + 4\sinh 3t\}$.

- b) Find $L\{e^{-3t}(\sin 2t + \cos 2t)\}$.

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4. a) Find $L\left\{\frac{\sin t}{t}\right\}$
b) Evaluate $L^{-1}\left\{\frac{5s^2 - 15s - 11}{(s+1)(s-2)^3}\right\}$
5. a) Solve $\frac{d^2y}{dt^2} - 2\frac{dy}{dt} + 2y = 0$, given $y(0) = y'(0) = 1$
b) Evaluate $L^{-1}\left\{\log\left(\frac{s+4}{s+3}\right)\right\}$
6. a) Find the analytic function of which the real part is
 $u = x^3 - 3xy^2 + 3x + 1$.
b) Evaluate the integral $\int_0^{1+i} z^2 dz$.
7. a) If $\vec{r} = 3\hat{i} - 6t^2\hat{j} + 4t\hat{k}$, find $\frac{d\vec{r}}{dt}$ and $\frac{d^2\vec{r}}{dt^2}$.
b) If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, show that $\text{grad } r = \hat{r}$.
8. a) Show that $\text{div } r^n \vec{r} = (n+3)r^n$.
b) If $\vec{F} = 3xy\hat{i} - y^2\hat{j}$. Evaluate $\int_C \vec{F} \cdot d\vec{r}$, where C is the curve
 $y = 2x^2$ in the xy-plane from (0, 0) to (1, 2).

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